

MIPS Datapath Lab Report

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CS161 – Lab 2 and Lab 3

1 Type Decode Unit

Table 1: Opcode patterns for instruction types.

Input (opcode bit)	R-format	lw	sw	beq
Op5	0	1	1	0
Op4	0	0	0	0
Op3	0	0	1	0
Op2	0	0	0	1
Op1	0	1	1	0
Op0	0	1	1	0

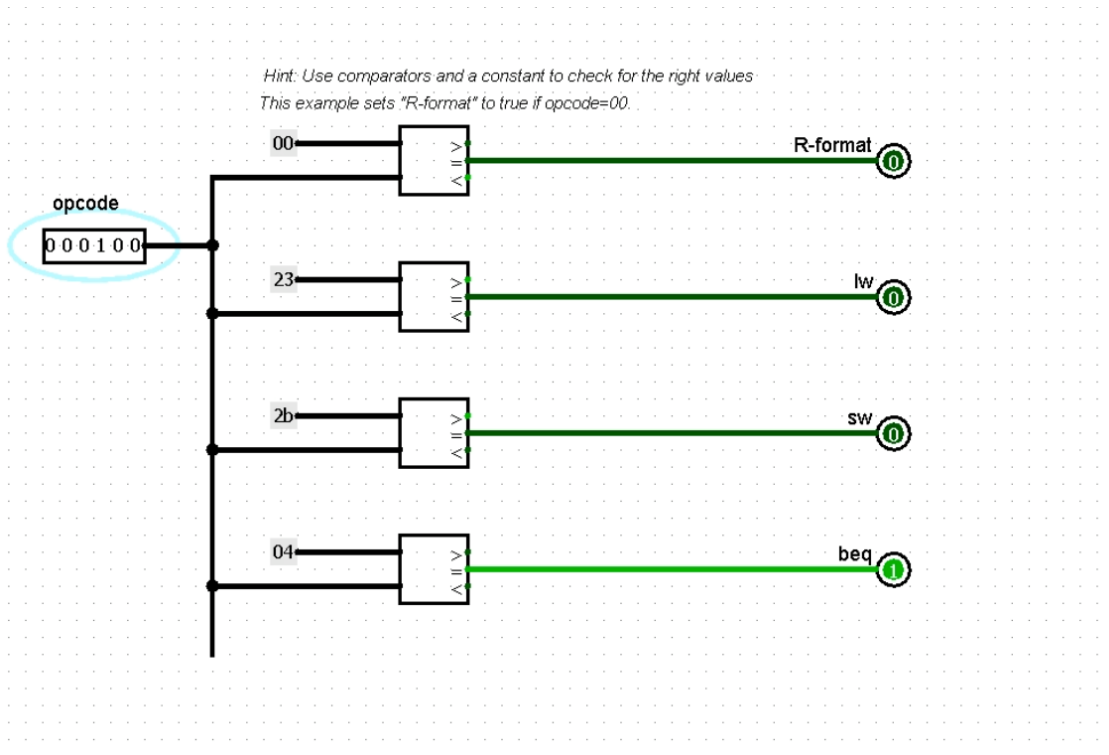


Figure 1: Type Decode unit after adding opcode comparators.

2 Control Decode Truth Table

Table 2: Control-decode truth table.

Output	R-format	lw	sw	beq
RegDst	1	0	<i>X</i>	<i>X</i>
ALUSrc	0	1	1	0
MemtoReg	0	1	<i>X</i>	<i>X</i>
RegWrite	1	1	0	0
MemRead	0	1	0	0
MemWrite	0	0	1	0
Branch	0	0	0	1
ALUOp1	1	0	0	0
ALUOp0	0	0	0	1

3 Control Decode Implementation

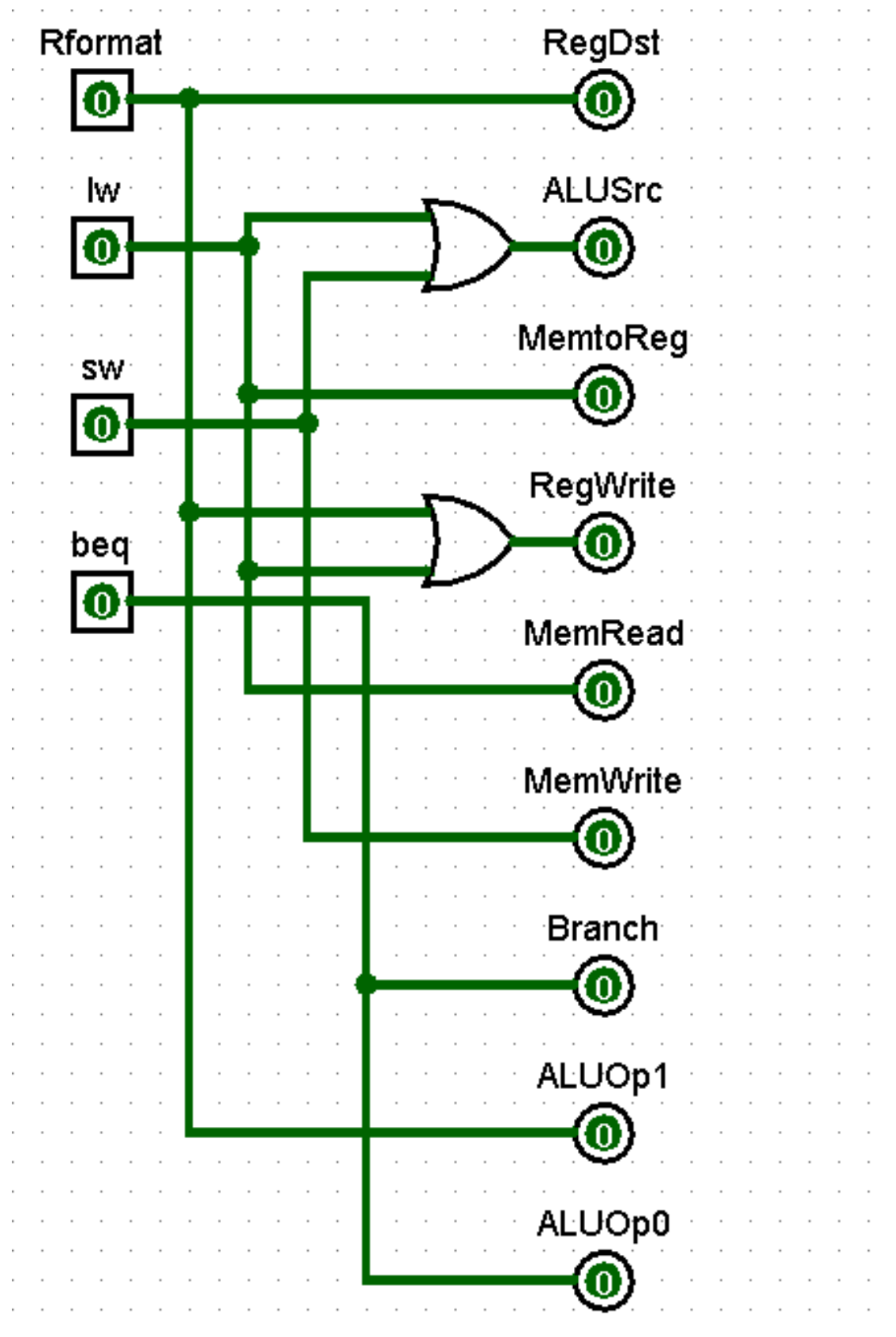


Figure 2: Control Decode unit with all outputs wired.

4 Control Logic Wiring

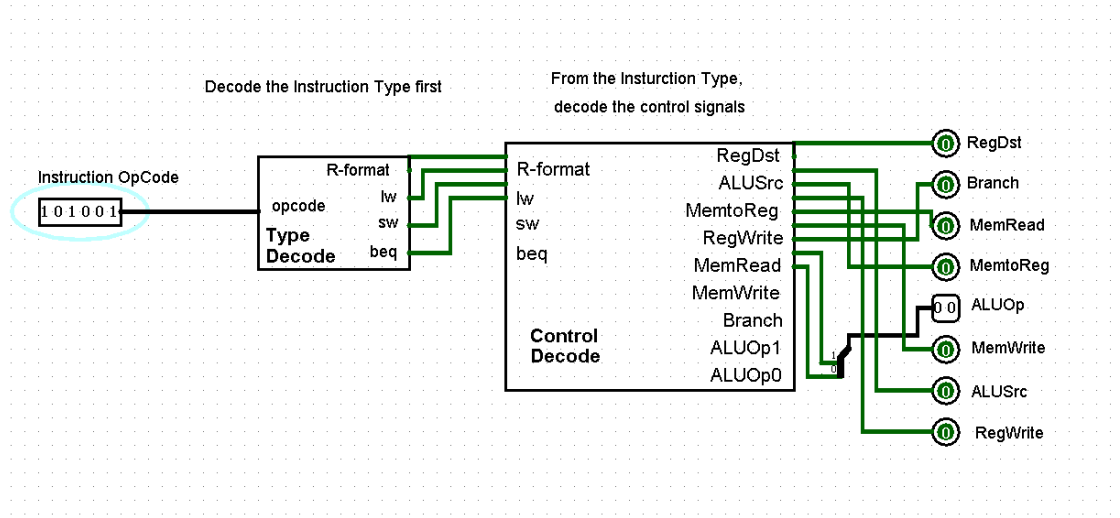


Figure 3: Completed control logic wiring.

5 ALU Control Unit

Table 3: ALU control mapping.

ALUOp1	ALUOp0	F5	F4	F3	F2	F1	F0	Operation
0	0	X	X	X	X	X	X	0010
0	1	X	X	X	X	X	X	0110
1	0	X	X	0	0	0	0	0010
1	0	X	X	0	0	1	0	0110
1	0	X	X	0	1	0	0	0000
1	0	X	X	0	1	0	1	0001
1	0	X	X	1	0	1	0	0111

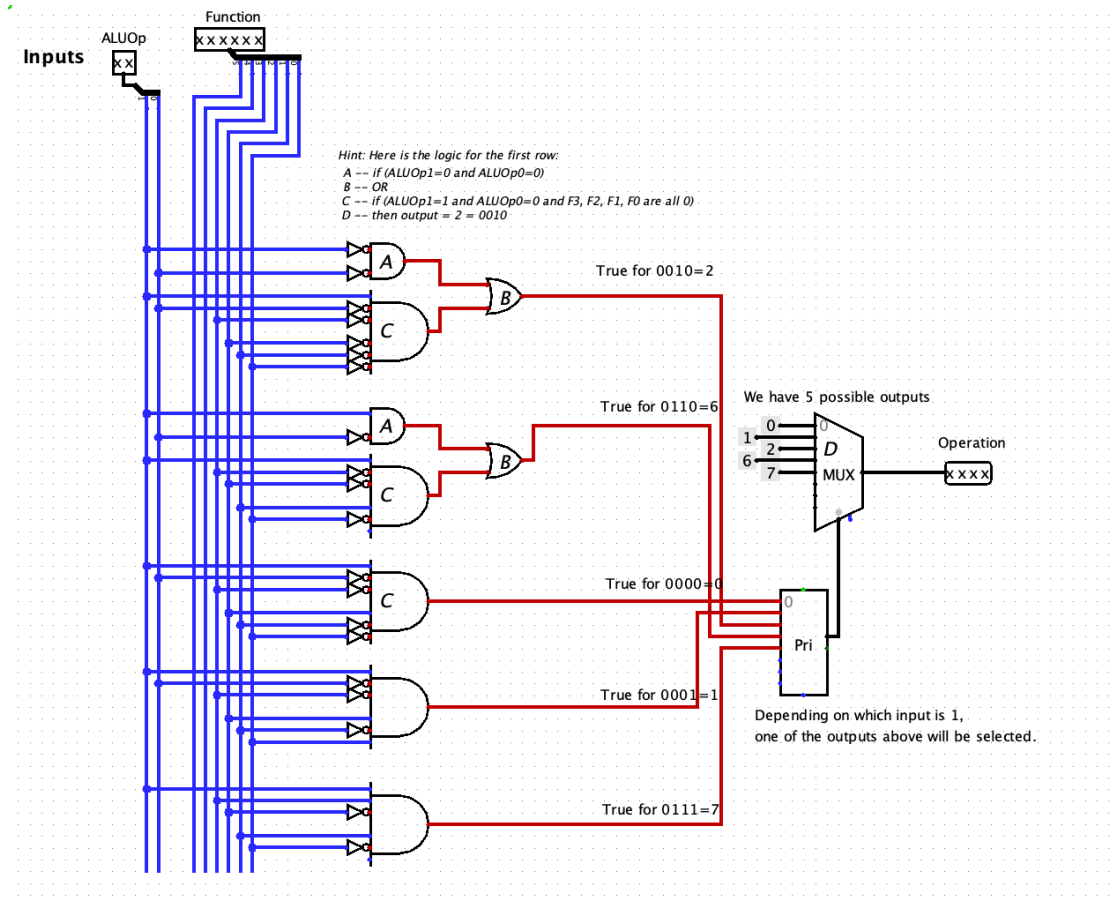


Figure 4: ALU Control with priority encoder and mux logic filled in.

6 Datapath Wiring

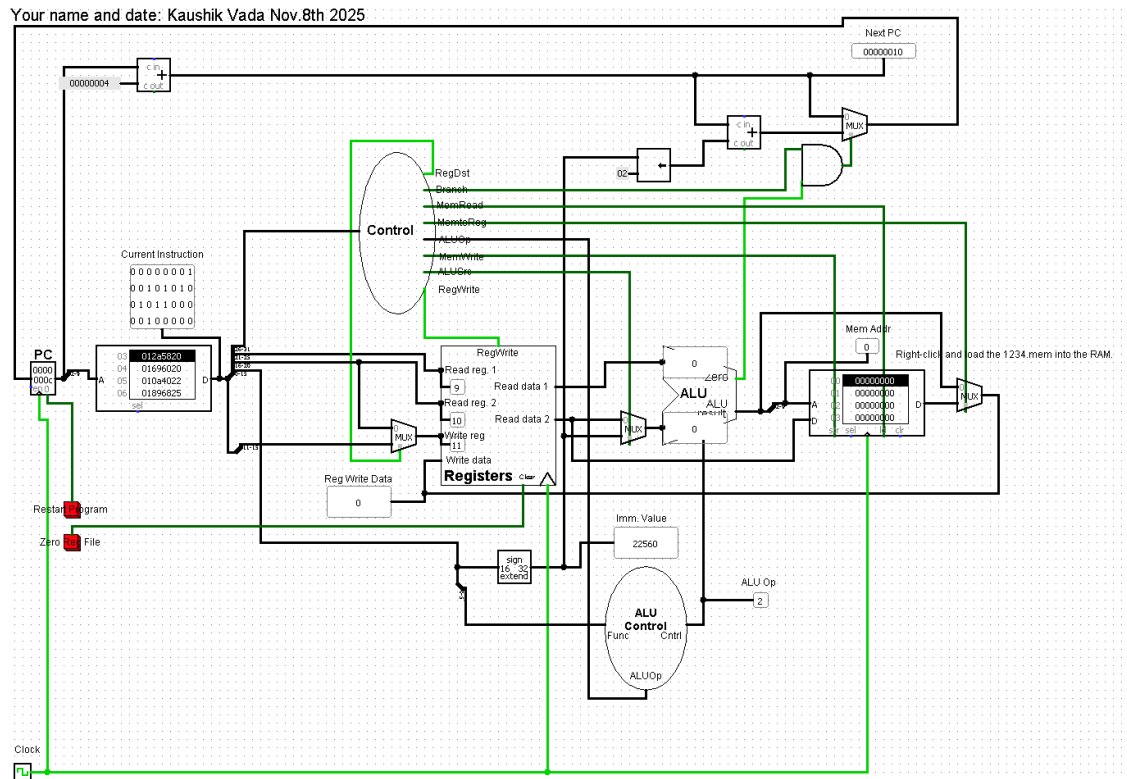


Figure 5: Completed processor wiring (datapath view).

7 Program Execution Results

Table 4: Final register file contents to report.

Register	$t0$	$t1$	$t2$	$t3$	$t4$	$t5$	$t6$
Value	0	1	2	3	4	5	1